

Uranus and Neptune

The two planets farthest from Earth, Uranus and Neptune are also the smallest of the gas giants. They are cold, featureless worlds, so remote that they are difficult to see from Earth. Both have sparse rings and large numbers of moons orbiting around them. And both are colored blue by the presence of methane-ice clouds in their atmospheres.

URANUS

The visible surface of Uranus is its hydrogen-rich atmosphere. Below is a deep layer of water, methane, and ammonia ices, and in the center a core of rock and possibly ice. Uranus appears calm and featureless because of upper-atmosphere haze, but the planet does undergo change. The atmosphere has some banding, and bright clouds are carried around the planet; in 2006, a dark spot was observed. Uranus's unique feature is its tilt: the planet appears to orbit the Sun on its side. The combination of the tilt and its 84-year orbit mean that its hemispheres face the Sun for 42 years at a time.

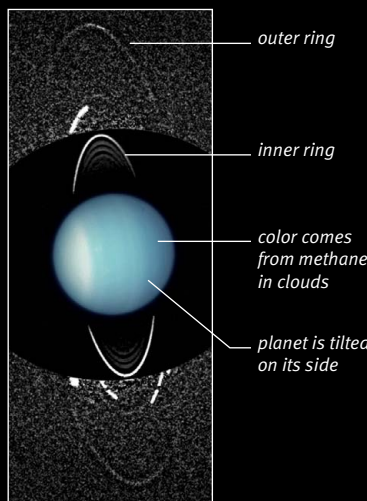


PLANET DATA

DIAMETER 31,763 miles (51,118 km)
AVERAGE DISTANCE FROM SUN 1.78 billion miles (2.87 billion km)
ORBITAL PERIOD 84 Earth years
ROTATION PERIOD 17.2 hours
CLOUD-TOP TEMPERATURE -353°F (-214°C)
NUMBER OF MOONS 27

Ring system

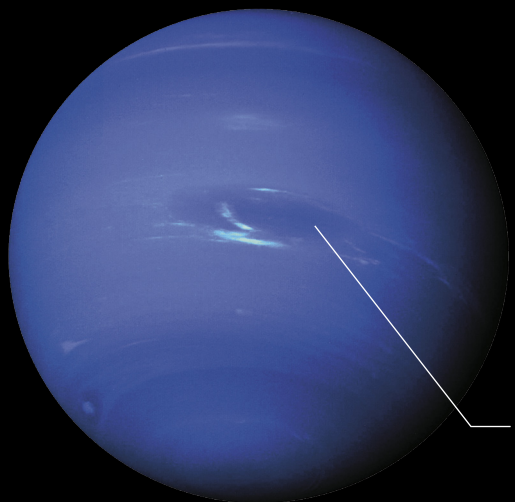
Uranus has two distinct sets of rings. Closest to the planet is a set of 11 rings; narrow and widely separated, these are more gap than ring. Outside these is a pair of faint, dusty rings.



NEPTUNE

Neptune is similar in structure to Uranus. It has a core of rock and possibly ice, surrounded by a mix of water, methane, and ammonia ices, and topped off with a hydrogen-rich atmosphere. The planet undergoes seasonal change, and its atmosphere is unexpectedly dynamic. It experiences ferocious equatorial winds, fast-moving bright clouds, and short-lived gigantic storms.

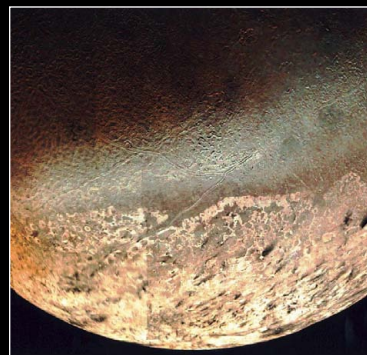
The planet is surrounded by a sparse system of five complete rings and one partial ring. Only one of its 13 moons, Triton, is of notable size, and four are within the planet's ring system.



the Great Dark Spot—a huge storm in the atmosphere

PLANET DATA

DIAMETER 30,760 miles (49,532 km)
AVERAGE DISTANCE FROM SUN 2.8 billion miles (4.5 billion km)
ORBITAL PERIOD 164.9 Earth years
ROTATION PERIOD 16.1 hours
CLOUD-TOP TEMPERATURE -320°F (-200°C)
NUMBER OF MOONS 13



Triton

Triton is about three-quarters the size of Earth's Moon. It is a rocky world with an icy surface of linear grooves, ridges, and circular depressions.

OBSERVING THE TWO PLANETS

Both Uranus and Neptune are too distant to be seen easily. The sharp-sighted can see Uranus with the naked eye, but it is difficult to locate. Following its progress against the stars will confirm its identity. This will take time and patience, because Uranus's orbit is long, and the planet spends about seven years in the same

zodiac constellation. Neptune moves even more slowly through the zodiac constellations, spending nearly 14 years in each. It is too faint to be seen by the naked eye, but a medium-sized telescope will reveal a blue-green disk.

Uranus and Neptune

Distant Uranus (left) is magnitude 5.5 and brighter than Neptune (right), which, at magnitude 7.8, is beyond naked-eye vision. Both appear as faint stars in this enhanced view.

